

IEORE 4004 - Quiz 1

This exam is composed of 10 pages and contains 2 exercises, plus a bonus exercise. The score for the bonus exercise will be computed separately and added as a bonus after the grades have been curved.

When solving an exercise, you can assume as known basic linear algebra facts, and all the statements seen in class and in the assignments, as well as closely related statements from the book and lecture notes. You cannot build on other results. If you have a doubt on whether you can use some result, ask us.

You cannot talk to your fellow students, for any reason. If you have a question, raise your hand and we will answer it. You are allowed to use one regular-sized cheatsheet, handwritten by yourself. Other than that, you are not allowed to use internet, books, notes,...

Motivate appropriately all your answers. There are many empty pages to use to write your solution, or as scratch paper. If you use all of them, you can request additional scratch paper. Submit everything, including the scratch paper and the cheatsheet.

Please do NOT open the exam until you are told to do so.

When you are done reading this page, write your name and uni below, then raise your hand.

Name: _____

UNI: _____

Good luck!

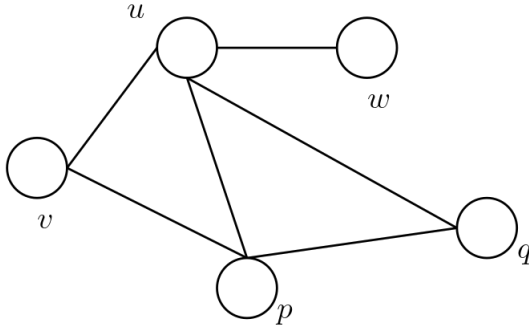
1. [50 points] Your company sells phones and owns one warehouse and one shop. In days 1,2,3, the shop needs to satisfy a given demand $d_1, d_2, d_3 > 0$ of phones, respectively. For $j \in \{1, 2, 3\}$, phones can be shipped from the warehouse to the shop in day j at a given per-unit cost $c_j > 0$. For $j \in \{1, 2\}$, phones already at the shop can be stored between day j and $j + 1$ at a given per-unit cost $q_j > 0$. Assume that the warehouse has enough capacity as to satisfy the demands of the shop.

1. Formulate the problem of shipping the phones from the warehouse to the shop in the different days as to satisfy the demand at minimum cost as an integer program.
2. Give an integer programming formulation for the following modification of the problem from part 1.: to use storage (in day 1 or day 2), you need to pay a one-time fee of k .
3. Give an integer programming formulation for the following modification of the problem from part 1.: We can decide to hire an external shipper for shipping goods in day 1. If we do so, we pay a fixed cost of ℓ , but the per-unit shipping cost c_1 is divided by 4. Else, we pay for shipping costs as in part 1.

2. [50 points] Recall that an *independent set* of a graph $G(V, E)$ is a set $S \subseteq V$ such that no two vertices of S share an edge – that is, there exists no $i, j \in S$ such that $ij \in E$. The Maximum Independent Set problem is the following:

- Given: A graph $G(V, E)$;
- Find: An independent set of G of maximum cardinality.

1. Formulate the Maximum Independent problem for the graph below as an integer program by explicitly listing the objective function and all the constraints.



2. Formulate the Maximum Independent Set problem for a generic graph as an integer program.
3. Recall that a *vertex cover* of a graph $G(V, E)$ is a set of nodes $S \subseteq V$ such that, for each edge $uv \in E$, $\{u, v\} \cap S \neq \emptyset$. That is, for every edge e of the graph, S contains at least one of the endpoints of the edge e . Explain why, if S is an independent set of a graph, $V \setminus S$ is a vertex cover.

Bonus question. Let $D(V, A)$ be a directed graph, $s \neq v \in V$, and $c : A \rightarrow \mathbb{N}$ a cost function on the arcs of D . Recall that from the optimal solution to the maximum $s - t$ flow problem, we can obtain an optimal solution to the minimum $s - t$ cut problem. Call $S \subseteq V$ be a *general cut* of D if $S, V \setminus S \neq \emptyset$. Give an efficient algorithm to find a general cut of minimum cost, where the cost of a general cut S is given by

$$\sum_{a=uv \in A: u \in S, v \notin S} c(a).$$